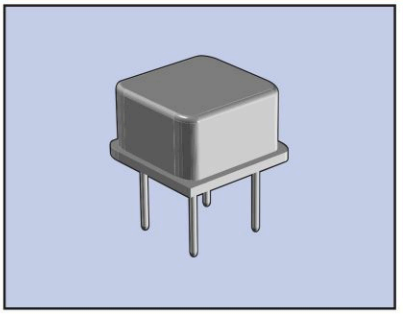
ECS-2100 SERIES 8 PIN DIP CLOCK OSCILLATOR



The ECS-2100 Series clock oscillator offers low current drain and is compatible with HCMOS/TTL logic. The metal package with pin #4 case ground acts as a shielding to minimize radiation.

FEATURES

- HCMOS/TTL logic compatible
- Wide frequency range
- Low power consumption
- Resistance weld package
- 3.3V operation (optional)

PART NUMBERING GUIDE

PART NUMBER *	FREQUENCY STABILITY
ECS-2100A	±100 PPM
ECS-2100B	±50 PPM
ECS-2100C	±25 PPM

** Complete part number to include frequency. i.e. ECS-2100A-100 (100 = 10.000MHz)*

OPERATING CONDITIONS/ELECTRICAL CHARACTERISTICS

PARAMETERS	FREQUENCY RANGE	CONDITIONS	MINIMUM	TYPICAL	MAXIMUM	UNITS
FREQUENCY RANGE (f_0)	1.000 ~ 150.000		1.000		150.000	MHz
OPERATING TEMP. RANGE (T_{OPR})	1.000 ~ 150.000		0		+70	˚C
STORAGE TEMP. RANGE (T_{STG})	1.000 ~ 150.000		-55		+125	˚C
FREQUENCY STABILITY	1.000 ~ 150.000	All conditions*	-100		+100	PPM
INPUT CURRENT (I_{DD})	1.000 ~ 25.000			17	25	mA
	25.000 ~ 50.000			33	46	mA
	50.000 ~ 80.000			45	77	mA
	80.000 ~ 150.000			67	82	mA
OUTPUT SYMMETRY	1.000 ~ 80.000	50% V_{DD} level	45	50 ±3	55	%
	80.000 ~ 150.000	50% V_{DD} level	40	50 ±3	60	%
RISE TIME (T_R)	1.000 ~ 150.000	10% ~ 90% V_{DD} level			5	nS
FALL TIME (T_F)	1.000 ~ 150.000	90% ~ 10% V_{DD} level			5	nS
OUTPUT VOLTAGE (V_{OL})	1.000 ~ 150.000	I_{OL} = 16 mA			0.5	V
OUTPUT VOLTAGE (V_{OH})	1.000 ~ 150.000	I_{OH} = -16 mA	4.5			V
OUTPUT CURRENT (I_{OL})	1.000 ~ 100.000	V_{OL} = 0.5 V			16	mA
OUTPUT CURRENT (I_{OH})	1.000 ~ 150.000	V_{OH} = 4.5 V			-16	mA
OUTPUT LOAD	1.000 ~ 150.000	TTL			10	TTL
	1.000 ~ 80.000	HCMOS			50	pF
	80.000 ~ 150.000	HCMOS			30	pF
START-UP TIME (T_S)	1.000 ~ 150.000	0.0V TO 5.0V			10	mS
SUPPLY VOLTAGE (V_{DC})		+5.0 ±0.25				V_{DC}

** Inclusive of 25˚C tolerance, operating temperature range, input voltage change, load change, aging, shock and vibration.*

PACKAGE DIMENSIONS (mm)

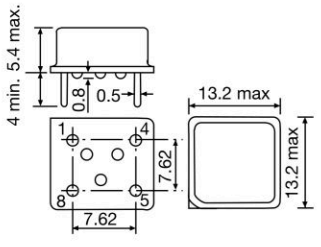


Figure 1) ECS-2100 Series – Side, Bottom and Top views

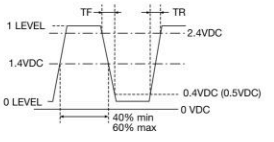


Figure 2) TTL Output Wave Form

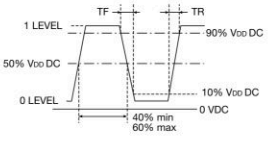


Figure 3) HCMOS Output Wave Form

PIN CONNECTIONS	
#1	NC
#4	GROUND
#5	OUTPUT
#8	+5V DC